Midterm Exam — Practice Form — Answer Key

INSTRUCTOR COPY — ECON 470

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Problem 1: Concepts [15 pts]

(a) [3 pts] Both microGPT and microOLS are estimating a *conditional distribution* and then sampling from it. microGPT predicts the next character given the preceding 16 (or 24) characters of context; microOLS predicts $\ln Q$ given $\ln P$ and the brand dummies. The analogue of "the next character" in microOLS is the next observation of $\ln Q$ — i.e., a draw of log-quantity given price and brand. The substrate (`Value`, `backward()`, Adam) does not care whether the conditional distribution is over characters or over log-sales; both are prediction problems in the Gans Ch. 1 sense.

(b) [3 pts] When $e = 1/2$, the signal fires at the same rate (1/2) regardless of the true state: $\Pr(s = 1 \mid \theta = 1) = \Pr(s = 1 \mid \theta = 2) = 1/2$. The joint table factors — s and θ are independent — so Bayes's rule gives a posterior equal to the prior. The signal carries no information; it cannot move beliefs and therefore cannot change the optimal action.

(c) [3 pts] Two acceptable comparative statics: - $\partial V/\partial e = R - r > 0$: better signals are more valuable. Each marginal point of accuracy is worth the full payoff gap $R - r$. - $\partial V/\partial(R - r) = e - p > 0$ (when $e > p$): higher-stakes decisions extract more value from the same signal. The signal's value scales with the size of the mistake it can avert. - (Also acceptable) $\partial V/\partial p < 0$ when $e > p$: as the prior gets stronger, the signal has less work to do.

(d) [3 pts] Without AI, the firm has *already* invested in protection x^*, which raises the probability of matching the state from p to $p + x^*$. To make adopting AI worthwhile, the signal must beat this *enhanced* prior, not the raw one. So $e > p$ is necessary but not sufficient; the right threshold is $e > p + x^*_{RULE}$.

(e) [3 pts] "Actionable" means the experimental result *changes what the agent does*. With prediction but no judgment, the agent doesn't know the payoff function, so even knowing θ doesn't tell them which action to take — by A1 they default to safe in both states. With judgment but no prediction, learning the payoff function is mostly inert (especially in the mixed case A2). Prediction converts mixed-case experimental outcomes from inert to action-changing — that's the complementarity.

Problem 2: 2$×$2 Prediction [30 pts]

Parameters: $p = 0.6$, $R = 80$, $r = 20$, $e = 0.8$.

Part A

(a) [3 pts] Joint table:

	$s = 1$	$s = 2$
$\theta = 1$	$ep = 0.48$	$(1-e)p = 0.12$
$\theta = 2$	$(1-e)(1-p) = 0.08$	$e(1-p) = 0.32$

Sum: $0.48 + 0.12 + 0.08 + 0.32 = 1.00$. ✓

(b) [3 pts]

$$\Pr(\theta = 1 \mid s = 1) = \frac{0.48}{0.48 + 0.08} = \frac{0.48}{0.56} = \frac{6}{7} \approx 0.857.$$

$$\Pr(\theta = 1 \mid s = 2) = \frac{0.12}{0.12 + 0.32} = \frac{0.12}{0.44} = \frac{3}{11} \approx 0.273.$$

(c) [3 pts] $0.857 > 0.6 = p$ and $0.273 < 0.6 = p$. The signal pushes the posterior toward $\theta = 1$ when $s = 1$ and toward $\theta = 2$ when $s = 2$ — that is what informative means: the signal moves beliefs in the *correct* direction.

Part B

(d) [3 pts] Since $p = 0.6 > 0.5$, the better rule is $a = 1$ (always prescribe):

$$E[U_{\text{rule}}] = pR + (1-p)r = 0.6(80) + 0.4(20) = 48 + 8 = 56.$$

(e) [6 pts] Strategy "follow the signal" $a = s$:

	$s = 1, a = 1$	$s = 2, a = 2$
$\theta = 1$	match: $R = 80$	mismatch: $r = 20$
$\theta = 2$	mismatch: $r = 20$	match: $R = 80$

$$E[U_{\text{follow}}] = 0.48(80) + 0.12(20) + 0.08(20) + 0.32(80) = 38.4 + 2.4 + 1.6 + 25.6 = 68.$$

(f) [3 pts]

$$V(e) = 68 - 56 = 12.$$

Check: $(e-p)(R-r) = (0.8-0.6)(80-20) = 0.2 \times 60 = 12$. ✓

Part C

(g) [3 pts] $\text{WTP}_i = (e-p)(R_i - r_i) = 0.2(R_i - r_i)$. - Lowest stakes: $0.2(20) = 4$. *No one* adopts when $P > 0.2 \times 120 = 24$. - Highest stakes: $0.2(120) = 24$. *Everyone* adopts when $P < 0.2 \times 20 = 4$.

So no one adopts at $P > 24$; everyone adopts at $P < 4$.

(h) [2 pts] Cutoff: $0.2(R_i - r_i) \geq 12 \implies R_i - r_i \geq 60$. Fraction with stakes ≥ 60 on $[20, 120]$ is $(120-60)/(120-20) = 60/100 = 0.6$. So $0.6 \times 500 = \mathbf{300}$ pediatricians adopt.

Part D

(i) [4 pts] - *Diagonal TPR = FPR:* The signal fires with the same frequency regardless of the true state, so it carries no information and cannot move the posterior. Receiving such a signal does not change beliefs and so cannot change the optimal action — useless. - *AI getting better:* The ROC frontier shifts upward and to the left, toward the corner $(0, 1)$. At every level of FPR, the agent can achieve a higher TPR; equivalently, the area under the ROC curve increases.

Problem 3: Substitutes for Prediction [30 pts]

Parameters: $R = 100$, $r = 20$, $p = 0.6$, $A = 1$, $B = 5$.

Part A: The rule

(a) [2 pts]

$$\pi(x, \Delta) = (p + x)R + (1 - p - x)(r + \Delta) - C(x) - c(\Delta).$$

(b) [3 pts] FOC w.r.t. x: $R - (r + \Delta) - C'(x) = 0$, i.e.,

$$R - r - \Delta = \frac{1}{(0.4 - x)^2}. \qquad (*)$$

FOC w.r.t. Δ: $(1 - p - x) - c'(\Delta) = 0$, i.e.,

$$1 - p - x = \frac{5}{80 - \Delta}. \qquad (**)$$

(c) [4 pts] Multiply $(*)$ by $(1 - p - x)$ and use $(**)$ to substitute $1 - p - x = B/(R - r - \Delta)$:

$$(R - r - \Delta)(1 - p - x) = B \implies B^2/(R - r - \Delta) = A \cdot ...$$

Cleaner: from $(*)$ and $(**)$,

$$(R - r - \Delta)(1 - p - x)^2 = A, \quad (R - r - \Delta)(1 - p - x) = B,$$

so dividing gives $1 - p - x = A/B = 1/5 = 0.2$, hence

$$\boxed{x^*_{RULE} = 0.4 - 0.2 = 0.2.}$$

Substitute into $(**)$: $0.2(80 - \Delta) = 5 \implies 80 - \Delta = 25 \implies \boxed{\Delta^*_{RULE} = 55.}$

(d) [3 pts] Compute the payoff: - $(p+x^*)R = 0.8 \times 100 = 80$. - $(1-p-x^*)(r+\Delta^*) = 0.2 \times 75 = 15$. - $C(0.2) = 1 \cdot [1/0.2 - 1/0.4] = 5 - 2.5 = 2.5$. - $c(55) = -5\ln(1 - 55/80) = -5\ln(25/80) = -5\ln(0.3125) \approx -5(-1.1632) = 5.816$.

$$\pi^*_{RULE} = 80 + 15 - 2.5 - 5.816 \approx \boxed{86.684.}$$

Part B: AI at $e = 0.75$, $\lambda = 2$

(e) [3 pts] $1 - e = 5/(80 - \Delta) \implies 0.25 = 5/(80 - \Delta) \implies 80 - \Delta = 20 \implies \boxed{\Delta^*_{DEC} = 60.}$

(f) [3 pts] $c(60) = -5\ln(1 - 60/80) = -5\ln(0.25) \approx -5(-1.3863) = 6.931$.

$$\pi^*_{DEC} = 0.75(100) + 0.25(80) - 6.931 - 2 = 75 + 20 - 6.931 - 2 \approx \boxed{86.069.}$$

(g) [2 pts] $86.069 < 86.684$. The firm should **not adopt**.

(h) [2 pts] $e = 0.75 < p + x^*_{RULE} = 0.8$. The condition does not hold. The firm has already raised its effective prior to 0.8 via protection; the signal has to beat that, not just the raw $p = 0.6$.

Part C: Better AI at $e = 0.9$, $\lambda = 2$

(i) [3 pts] $0.1 = 5/(80 - \Delta) \implies 80 - \Delta = 50 \implies \boxed{\Delta^*_{DEC} = 30.}$ $c(30) = -5\ln(1 - 30/80) = -5\ln(0.625) \approx -5(-0.4700) = 2.350$.

$$\pi^*_{DEC} = 0.9(100) + 0.1(50) - 2.350 - 2 = 90 + 5 - 2.350 - 2 \approx \boxed{90.650.}$$

(j) [3 pts] $90.650 > 86.684$, so the firm **adopts**. Payoff rises by $90.650 - 86.684 \approx 3.97$.

(k) [2 pts] $\Delta^*_{DEC} = 30 < 55 = \Delta^*_{RULE}$: insurance **fell**. Consistent with Proposition 4.1: when $e > p + x^*_{RULE}$, the signal is accurate enough to reduce the firm's exposure to mismatch errors, so it needs less insurance.

Problem 4: Complements to Prediction [25 pts]

Parameters: $R = 100$, $r = 20$, $S = 70$, $v = 0.4$, $\delta = 0.9$.

Part A: Assumptions

(a) [3 pts] A1: $vR + (1-v)r = 0.4(100) + 0.6(20) = 40 + 12 = 52 \leq 70 = S$. ✓ Without A1, the risky action would beat safe even on the prior alone, so the agent doesn't need any information to act — and the whole "prediction makes experimentation actionable" story collapses.

(b) [3 pts] A2: $\frac{1}{2}(R + r) = \frac{1}{2}(120) = 60 \leq 70 = S$. ✓ A2 says: knowing only the *payoff function* — that one state pays R and the other r, but not which is which — is not enough to deviate from safe. The mixed-case experimental outcome is inert without prediction.

Part B: The four cells

(c) [3 pts] - $V_\emptyset = S = 70$. - $V_P = S = 70$ (prediction alone is inert under A1). - $V_J = v^2R + (1 - v^2)S = 0.16(100) + 0.84(70) = 16 + 58.8 = 74.8$. - $V_{PJ} = vR + (1-v)S = 0.4(100) + 0.6(70) = 40 + 42 = 82$.

(d) [3 pts] - $V_P - V_\emptyset = 0$. - $V_J - V_\emptyset = 74.8 - 70 = 4.8$. - $V_{PJ} - V_\emptyset = 82 - 70 = 12$. - Complementarity: $12 - 4.8 - 0 = 7.2$. Share: $7.2/12 = 60\%$.

The bulk of the joint surplus is interactive — it exists only when prediction and judgment are both present.

(e) [3 pts] Without judgment, the agent doesn't know what risky pays in either state. By A1, safe beats risky on the prior alone, so even knowing θ doesn't change the action: the agent picks safe regardless. Prediction adds no actionable information by itself; it only earns its keep when paired with judgment that has discovered the payoff function.

Part C: Threshold

(f) [3 pts]

$$\bar{\lambda} = \frac{70(0.1)}{0.9(0.4)(30)} = \frac{7}{10.8} \approx \boxed{0.648.}$$

(g) [2 pts] With $\delta = 0.99$:

$$\bar{\lambda} = \frac{70(0.01)}{0.99(0.4)(30)} = \frac{0.7}{11.88} \approx 0.0589.$$

The threshold **falls sharply** (from ≈ 0.648 to ≈ 0.059). A more patient agent loses very little by waiting one period to learn, so even very low-quality experiments are worth running.

(h) [2 pts] With $R = 130$, $R - S = 60$:

$$\bar{\lambda} = \frac{70(0.1)}{0.9(0.4)(60)} = \frac{7}{21.6} \approx 0.324.$$

The threshold falls. **Firms with a large $R - S$ have a stronger incentive to invest in prediction** — the upside per successful experiment is wider, so the same quality of signal generates more expected value.

(i) [3 pts]

$$\bar{\lambda}_{\text{no pred}} = \frac{70(0.1)}{0.9(0.16)(30)} = \frac{7}{4.32} \approx 1.620.$$

This exceeds 1, so no feasible signal quality $\lambda \in (0, 1)$ clears the bar. Without prediction, the agent **never experiments** at these parameters — prediction is *necessary*, not merely helpful, for experimentation to be worthwhile.

End of answer key.